

Exercise Sheet 2: Investment and Asset Prices

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Class plan. Groups 1–4 work on Problems 1–4 respectively. Each group prepares a short solution and a short economic interpretation. Problem 5 is the joint problem: start it in groups of 3–4, then solve and discuss it together with the TA. The optional extension is harder and can be used by groups that finish early or for additional discussion.

Problem 1 (Group 1). Asset prices and valuation

- (a) A one-period bond pays 1 next period with certainty. Let the real interest rate be r . Derive the bond price

$$P_t^B = \frac{1}{1+r}.$$

- (b) A project pays 30 next year, 40 in two years, and 60 in three years. Compute its present value when $r = 0.04$ and when $r = 0.07$. Which payment is most affected by the rise in r , and why?
- (c) A share costs V_t today and is expected to deliver dividend D_t^e plus expected resale value V_{t+1}^e next period. Investors require gross return $1 + r + \varepsilon$. Derive the one-period pricing equation for the share.
- (d) Suppose expected dividends grow at constant rate g forever, so that the Gordon formula applies:

$$V_t = \frac{D_t^e}{r + \varepsilon - g}.$$

Compute V_t when $D_t^e = 6$, $r = 0.03$, $\varepsilon = 0.04$, and $g = 0.02$. Then recompute V_t when only the real interest rate changes to $r = 0.05$.

- (e) Invert the Gordon formula. Suppose the observed dividend yield is $D_t^e/V_t = 0.015$, with $r = 0.03$ and $\varepsilon = 0.04$. What dividend growth rate g is needed to justify the observed valuation? Compare your answer to a long-run real GDP growth rate of 2.5% and interpret briefly.

Problem 2 (Group 2). Tobin's q and business investment with depreciation

Let capital accumulate according to

$$K_{t+1} = (1 - \delta)K_t + I_t,$$

where I_t is *gross* investment. Assume adjustment costs are

$$c(I_t) = \frac{a}{2}(I_t - \delta K_t)^2, \quad a > 0,$$

so replacement investment has zero adjustment cost, while deviations from replacement are costly. Replacement investment still has the direct purchase cost I_t . The firm chooses I_t to maximize

$$\Pi_t - I_t - c(I_t) + q_t((1 - \delta)K_t + I_t).$$

- (a) Differentiate with respect to I_t and derive the first-order condition

$$q_t = 1 + a(I_t - \delta K_t).$$

- (b) Solve explicitly for gross investment I_t . Interpret the two terms in your expression.
 (c) Show that in a steady state with $K_{t+1} = K_t$, gross investment satisfies

$$I^* = \delta K^*, \quad q^* = 1.$$

Why is $q = 1$ natural in the long run?

- (d) Numerical application: let $K_0 = 100$, $\delta = 0.05$, $a = 0.5$. Suppose $q_0 = 1.2$ in period 0 and $q_1 = 1.1$ in period 1. Compute I_0 , K_1 , I_1 , and K_2 .
 (e) Investment responds immediately to a change in q , but the capital stock responds only gradually. Explain why in one or two sentences.

Problem 3 (Group 3). Average q , marginal q , and investment subsidies

Let the value of the firm be

$$V(K) = AK^\alpha, \quad 0 < \alpha < 1,$$

with $A = 4$ and $\alpha = 0.8$. The firm faces quadratic adjustment costs

$$c(I) = \frac{a}{2}I^2, \quad a = 0.1.$$

Suppose a subsidy at rate s reduces the purchase price of capital from 1 to $1 - s$. In parts (c)–(d), treat q_t^m as evaluated at the current capital stock and taken as given when choosing current investment. This is the local q -theory approximation.

- (a) Derive average Tobin's $q^{avg} = V(K)/K$ and marginal Tobin's $q^m = V'(K)$. Show that

$$q^m = \alpha q^{avg}.$$

- (b) Compute q^{avg} and q^m at $K = 100$. By how much do they differ?
 (c) Using marginal q as the relevant incentive for investment, show that with a subsidy s the first-order condition becomes

$$q_t^m = (1 - s) + aI_t,$$

and solve for I_t .

- (d) Using the q^m from part (b), compute investment when $s = 0$ and when $s = 0.10$. How much does the subsidy raise investment?
 (e) Why can measured average q be a poor predictor of firm-level investment even when the q -theory is correct? Mention one role for decreasing returns and one role for policy.

Problem 4 (Group 4). Housing, taxation, and construction

The user cost of housing is

$$uc = [r(1 - \tau) + \delta + s]p_t^H \equiv v p_t^H, \quad v = r(1 - \tau) + \delta + s,$$

where τ captures the tax treatment of interest income or mortgage-interest deductibility, and s is a property-tax rate. For this exercise, ignore expected house-price gains or losses. Housing demand is

$$H_t^d = \frac{\eta Y_t}{v p_t^H},$$

housing supply is

$$I_t^H = k \left(\frac{p_t^H}{P_t} - 1 \right) + \delta H_t,$$

and the housing stock evolves according to

$$H_{t+1} = (1 - \delta)H_t + I_t^H.$$

Assume $P_t = P$ is constant.

- (a) Explain each term in $v = r(1 - \tau) + \delta + s$ in one sentence. Why does a lower value of τ raise the user cost of housing? Distinguish between an owner who finances housing from accumulated wealth and an owner with a mortgage.
- (b) In the short run, the stock H_t is predetermined. Show that short-run equilibrium in the housing market implies

$$p_t^H = \frac{\eta Y_t}{v H_t}.$$

Then combine this with supply and accumulation to show that

$$H_{t+1} = H_t + k \left(\frac{\eta Y_t}{v P H_t} - 1 \right).$$

- (c) Derive the long-run steady-state values

$$H^* = \frac{\eta Y}{v P}, \quad p^{H*} = P, \quad I^{H*} = \delta H^*.$$

Explain briefly why $p^{H*} = P$.

- (d) Numerical baseline: let $\eta = 0.2$, $Y = 100$, $P = 2$, $r = 0.03$, $\tau = 1/3$, $\delta = 0.025$, $s = 0.005$, and $k = 10$. Compute v , H^* , p^{H*} , and I^{H*} .
- (e) Now raise the property tax from $s = 0.005$ to $s = 0.015$, keeping everything else fixed. Compute the new long-run housing stock $H^{*'}$ and long-run investment $I^{H*'}$. Then compute the *immediate* short-run house price, housing q , and housing investment when the housing stock is still at its old steady-state level $H = 200$. Interpret the short-run and long-run effects in a few lines.

Problem 5 (joint problem). A discount-rate and user-cost shock: business vs housing investment

Start this problem in groups of 3-4. The solution and main discussion are intended to be done together with the TA.

A representative firm has current expected dividend $D_t^e = 5$, capital stock $K_t = 100$, $\delta = 0.05$, and adjustment-cost parameter $a = 0.5$. Assume the Gordon formula

$$V_t = \frac{D_t^e}{r + \varepsilon - g}$$

with $\varepsilon = 0.03$ and $g = 0.02$, and use the business-investment rule from Problem 2:

$$I_t = \delta K_t + \frac{q_t - 1}{a}, \quad q_t = \frac{V_t}{K_t}.$$

For housing, use the baseline from Problem 4:

$$H_t = 200, \quad P = 2, \quad \eta = 0.2, \quad Y = 100, \quad v = 0.05, \quad k = 10, \quad \delta = 0.025.$$

Suppose monetary easing lowers the relevant discount rate for the firm: the real interest rate falls from $r = 0.04$ to $r = 0.03$. For housing, assume directly that the user-cost rate falls from $v = 0.05$ to $v = 0.04$.

- Compute the firm's market value V_t , Tobin's q_t , and gross investment I_t *before* and *after* the fall in r .
- Compute the house price p_t^H , housing $q_t^H = p_t^H / P$, and housing investment I_t^H *before* and *after* the fall in v .
- Which type of investment responds more in your calculations? Compare both the level change and the percentage change.
- Give two economic reasons why business and housing investment may respond differently in practice, even when both are driven by asset prices.

Optional extension (harder). Tax change during a recession

Start from the baseline steady state of Problem 4 with $v = 0.05$, $Y = 100$, and $H = 200$. In period 1, two changes happen simultaneously: a property-tax increase raises v to 0.06, and a recession lowers aggregate income to $Y = 95$ for three periods. From period 4 onward, income returns to $Y = 100$.

- Compute the new long-run values under $v = 0.06$ and $Y = 100$.
- Compute p_1^H , q_1^H , I_1^H , and H_2 in the first period of the joint shock.
- Decompose the immediate fall in p^H using this ordering: first change v from 0.05 to 0.06 while holding $Y = 100$ and $H = 200$ fixed; then change Y from 100 to 95 while holding $v = 0.06$ and $H = 200$ fixed. What fraction of the total fall is due to the tax/user-cost change, and what fraction is due to the income shock?
- A policymaker argues that the tax change should have been deferred until the recession was over. Use the model to evaluate this argument in two or three sentences.
- Name one important real-world channel that is not captured by this model.